THOMAS MORE

Data Science Fundamentals

Logistic Regression

THOMAS MORE

Before we get started…

Practicalities

Learning objectives

- You understand and apply different types of regression, know when and how to use them, can execute them and put them in to practice as well as interpret the results.
- You recognize overfitting in regression models and learn techniques to prevent overfitting.
- You learn about different clustering techniques and understand when and how to use them for grouping data.
- You gain an understanding of dimensionality reduction techniques and learn how to apply them to compress and visualize data.

Course material

- **Canvas**
  - Slides & exercises from classes
- **Online**
  - https://www.w3schools.com/datascience/
- **Software**
  - JupyterLab (or work via Anaconda/VS Code if you prefer)
- **The Internet**



Course breakdown

- **Course Structure:**
  - Lectures: 3 hours per week (theory and exercises)
  - Attendance: Mostly not compulsory, but strongly recommended.

- **Examination Breakdown:**
  - 30%: Continuous assessment (mandatory excercises)
  - 70%: Final exam (oral, mostly theoretical with some practical elements)

Is it difficult to pass?

It’s perfectly doable **if you put in the work**

- Come to class
- Participate in the lessons
- Make your exercises



How to be an ideal student

- **Attend classes** - even when you party hardy
- **Show respect** - towards your instructor and fellow students
- **Be punctual** - or you'll find the door closed
- **Meet deadlines** - late submissions will earn a solid zero
- **Have a question?** - just ask! I don't bite
- **Know it all?** - help your classmates, they might repay you with pizza
- **Do your own work!** - shortcuts will lead to a short academic career ;)

Agenda

Week - 1	Introduction & Python/JupyterLab Setup	Week - 9	Easter Holiday
Week - 2	Regression - Linear regression	Week - 10	Easter Holiday
Week - 3	Regression - Polynomial regression 	Week - 11	Easter Monday
Week - 4	Classification - Logistic regression	Week - 12	Dimensionality reduction: Feature Analysis - PCA
Week - 5	Classification - Decision trees	Week - 13	Regularization: Lasso regression 
Week - 6	Classification - kNN 	Week - 14	Recap: Theory + Excercises
Week - 7	Clustering - kMeans	Week - 15	Exam Preperation + Q&A
Week - 8	Clustering - Hierarchical clustering		

Syllabus

Week - 1	Introduction & Python/JupyterLab Setup	Week - 9	Easter Holiday
Week - 2	Regression - Linear regression	Week - 10	Easter Holiday
Week - 3	Regression - Polynomial regression	Week - 11	Clustering - Hierarchical clustering
Week - 4	Classification - Logistic regression	Week - 12	Dimensionality reduction: Feature Analysis - PCA
Week - 5	Classification - Decision trees	Week - 13	Regularization: Lasso regression
Week - 6	Classification - kNN	Week - 14	Recap: Theory + Excercises
Week - 7	Clustering - kMeans	Week - 15	Exam Preperation + Q&A
Week - 8	Lifesnaps guidance		

(subject to change)

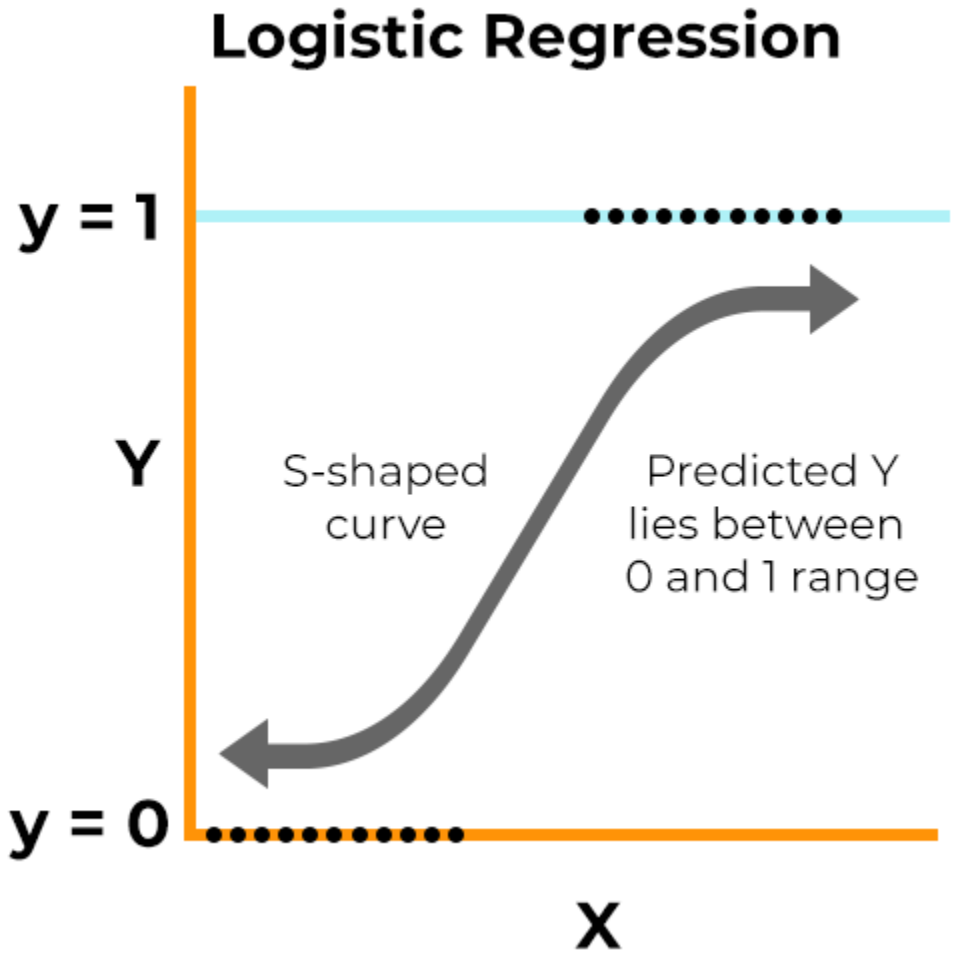
THOMAS MORE

Now we get to the fun part

Logistic Regression

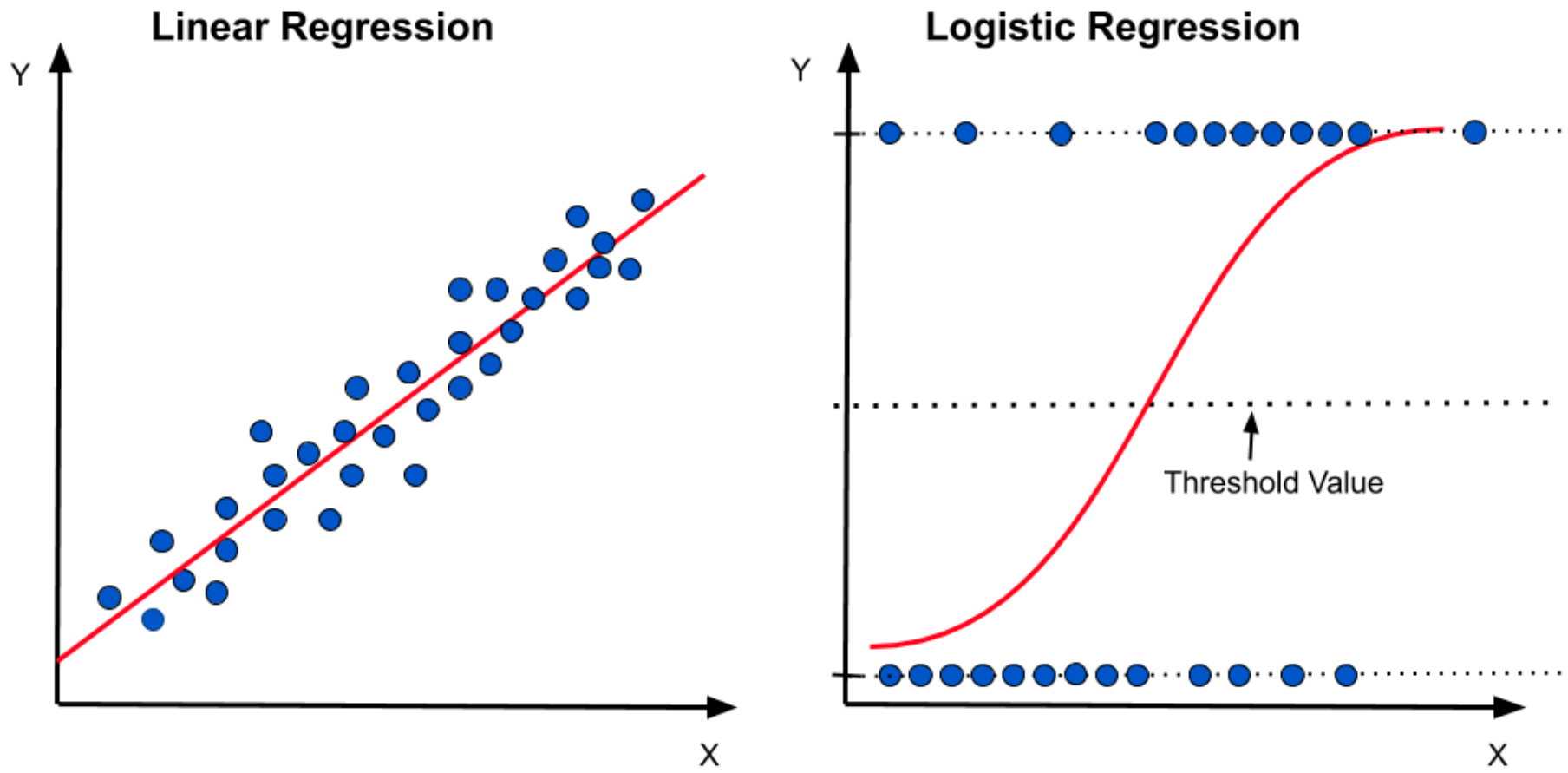
Regression, but not really

- In its core a form of regression
- **S-shaped** curve
- Based on X, we predict Y
- Y always lies between 0 and 1
  - This interesting property allows us to go beyond regression…



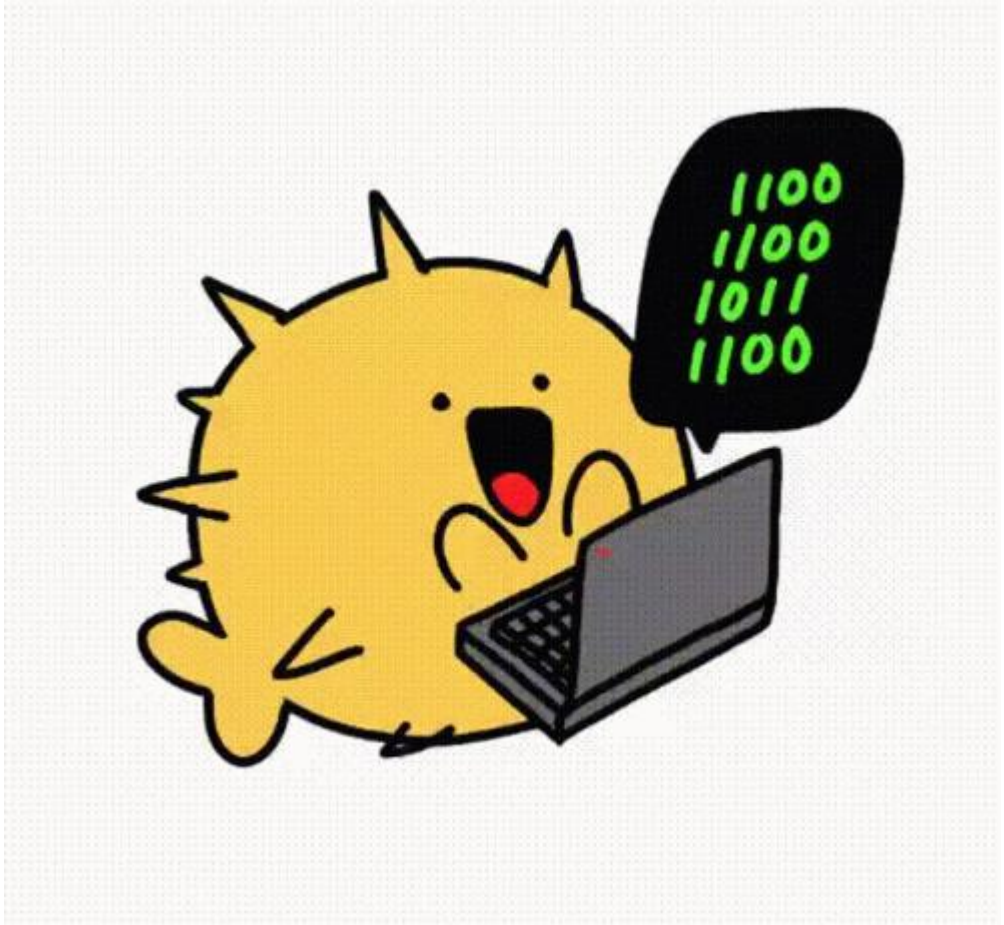
Regression as classification

*By establishing a '**threshold**' we can label each prediction as 0 or 1 → Classification!*



Binary classification

- Always **0** or **1**
- Predicts if a certain factor is **True** or **False**
  - E.g. Spam, Cancer, …
- Simple yet limited



Multiclass classification

- Can be anything
- Predicts the type of a certain factor
  - E.g. Wine quality rating, …
- More complicated yet (almost) unlimited

Is classification supervised or unsupervised?

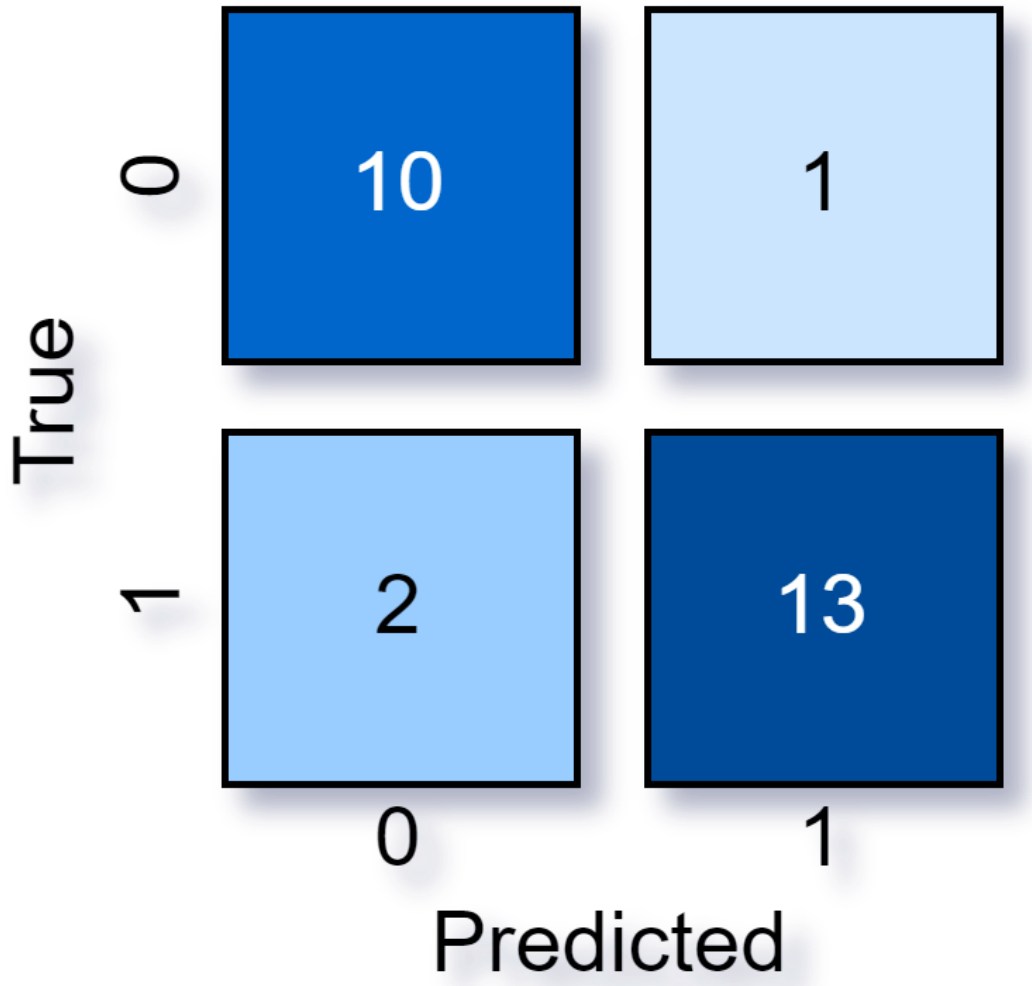
Supervised!

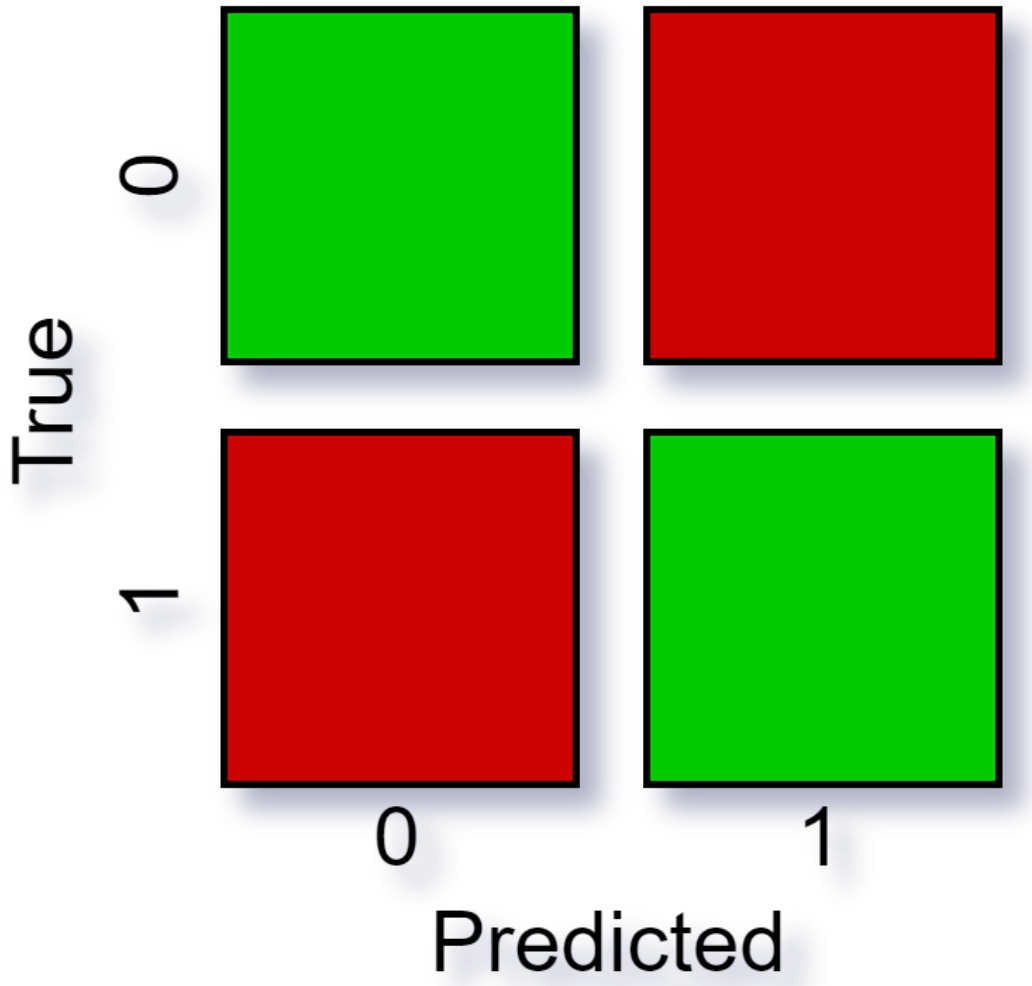


Evaluation metrics

- **True Positives (TP)**: The amount of labels where the outcome is 1 and the predicted label is 1. In other words, the amount of variables where we *correctly* predicted the postive outcome.
- **True Negatives (TN)**: The amount of labels where the outcome is 0 and the predicted label is 0. In other words, the amount of variables where we *correctly* predicted the negative outcome.
- **False Positives (FP)**: The amount of labels where the outcome is 0 and the predicted label is 1. In other words, the amount of variables where we *incorrectly* predicted the postive outcome.
- **False Negatives (FN)**: The amount of labels where the outcome is 1 and the predicted label is 0. In other words, the amount of variables where we *incorrectly* predicted the negative outcome.

Confusion matrix

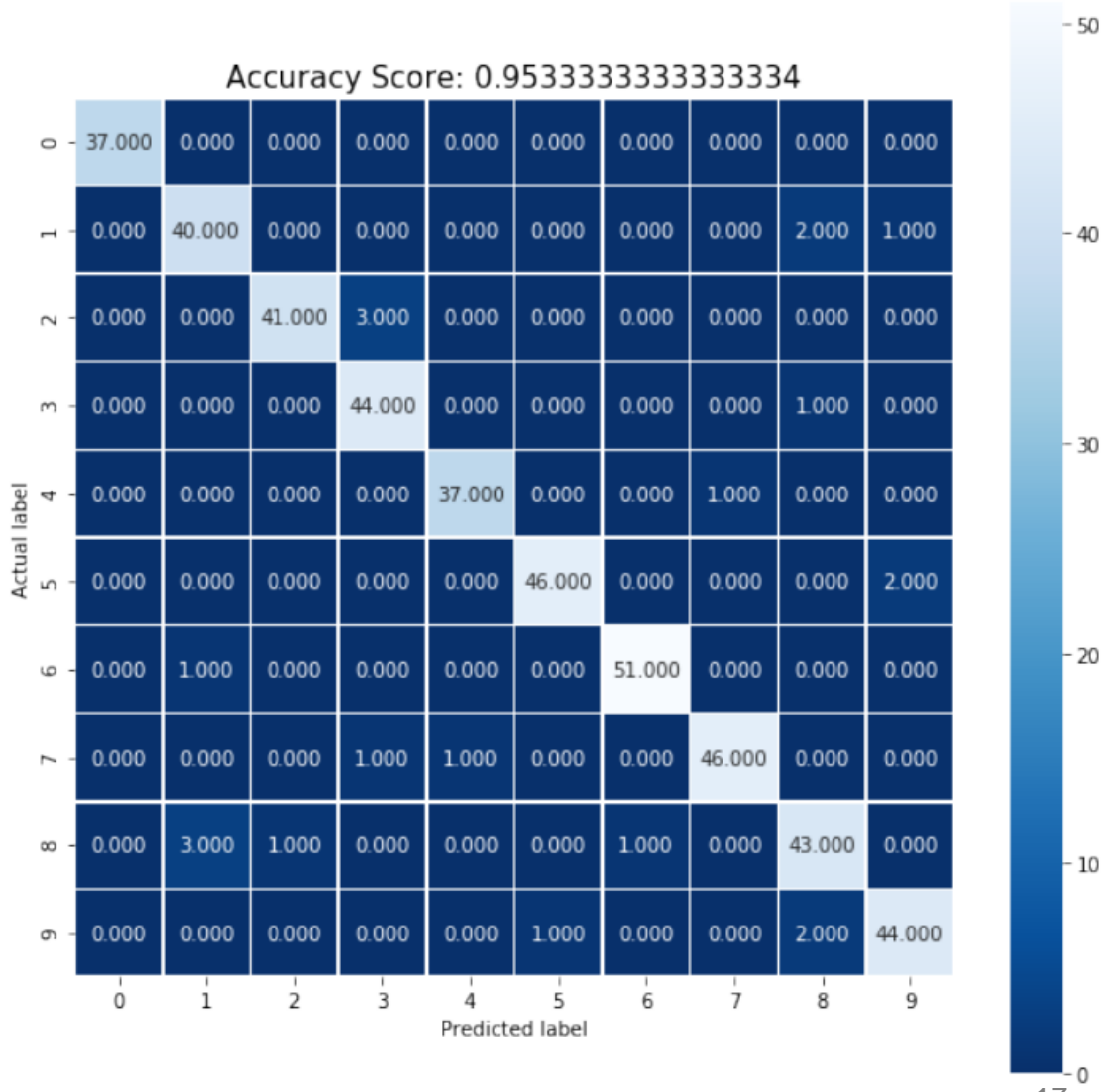




Confusion matrix

- The same goes for multiclass classification
- Calculate TP/TN/FP/FN **per class**
- You create an overal idea of your model by calculating an average
- Python can do both for us :D

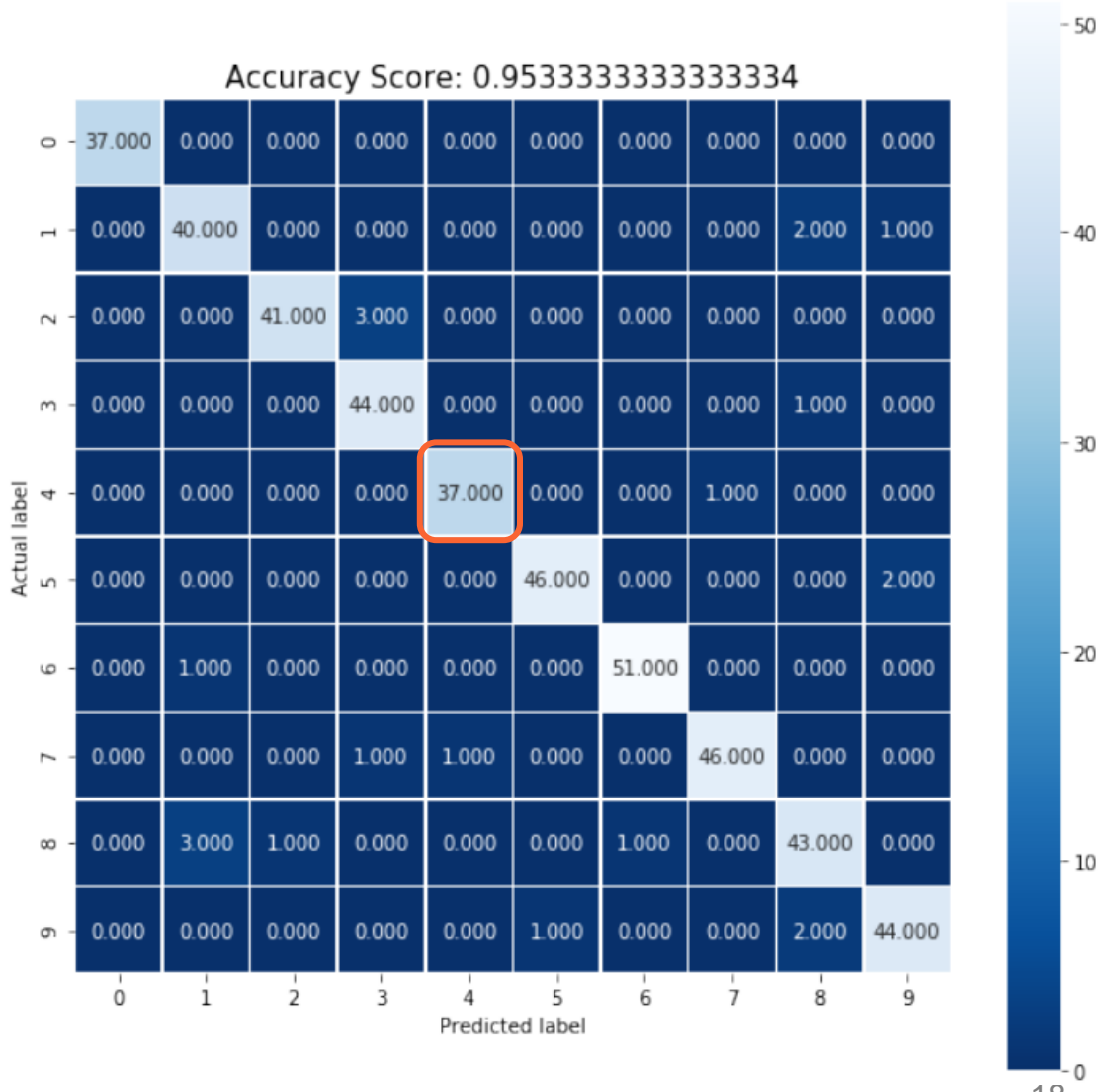
Now that we know the terminology, let's see what we can calculate with this…



Confusion matrix

- The same goes for multiclass classification
- Calculate TP/TN/FP/FN **per class**
- You create an overal idea of your model by calculating an average
- Python can do both for us :D

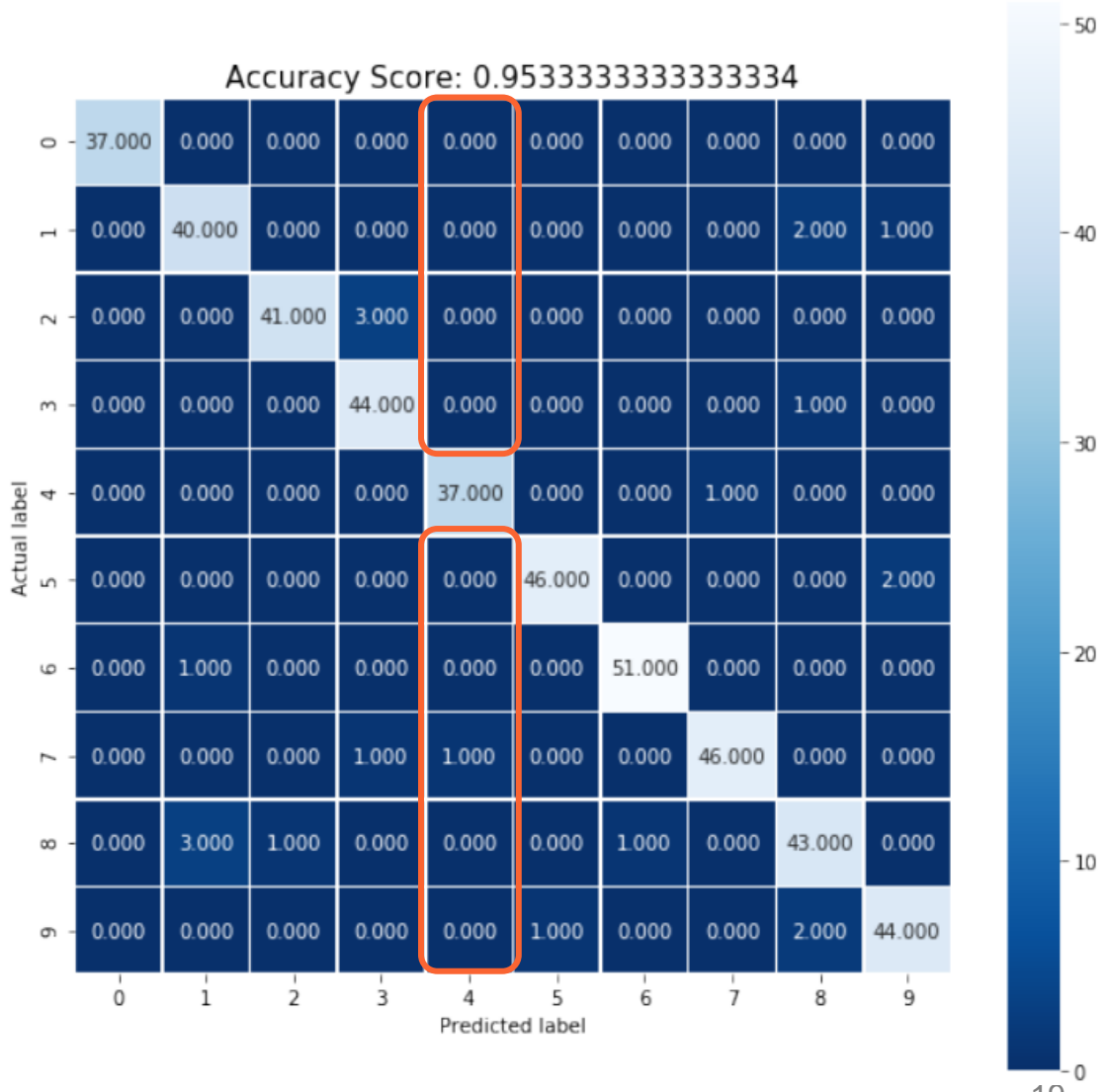
Example: class ‘4’
Red indicates TP



Confusion matrix

- The same goes for multiclass classification
- Calculate TP/TN/FP/FN **per class**
- You create an overal idea of your model by calculating an average
- Python can do both for us :D

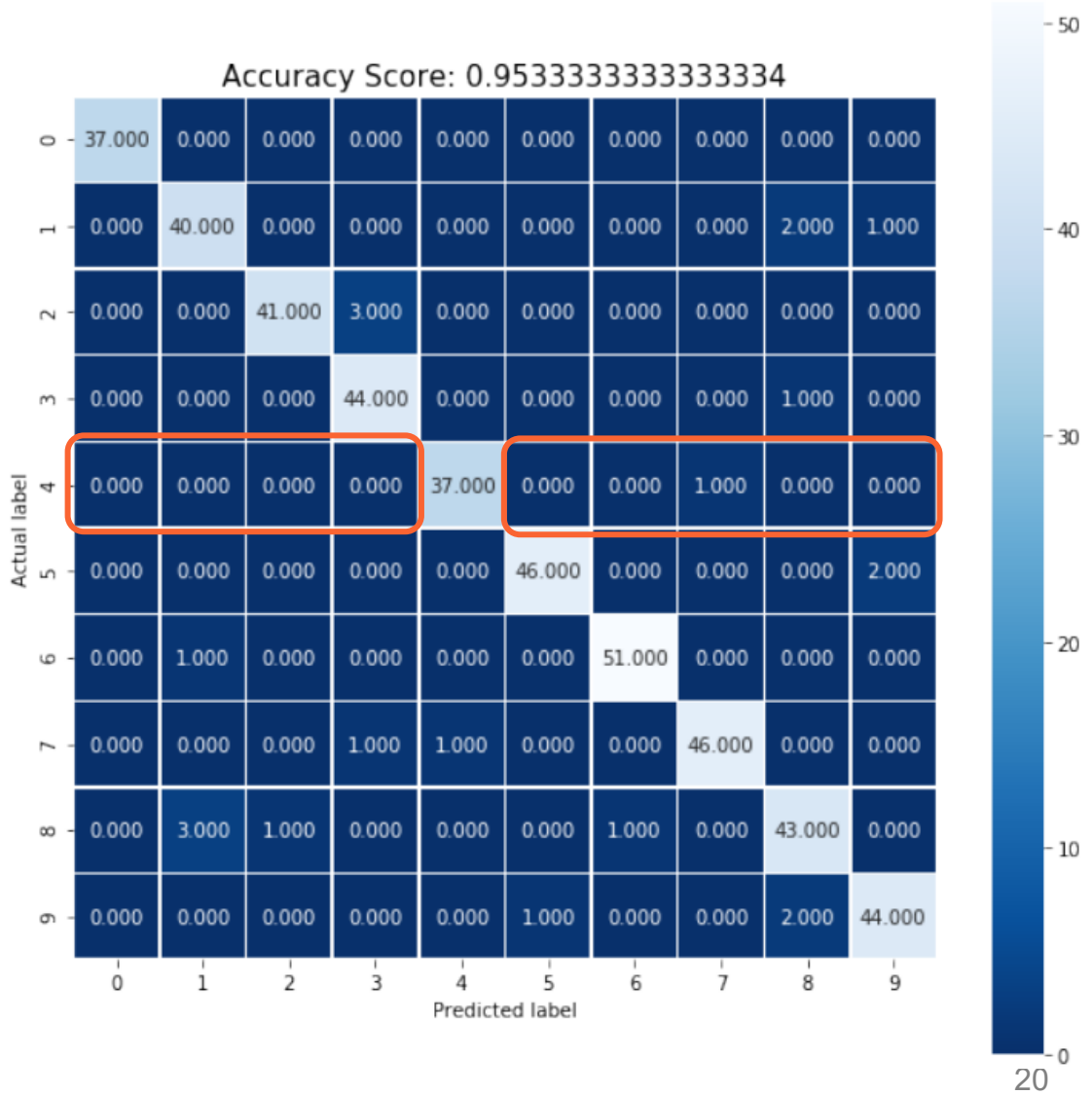
Example: class ‘4’
Red indicates FP



Confusion matrix

- The same goes for multiclass classification
- Calculate TP/TN/FP/FN **per class**
- You create an overal idea of your model by calculating an average
- Python can do both for us :D

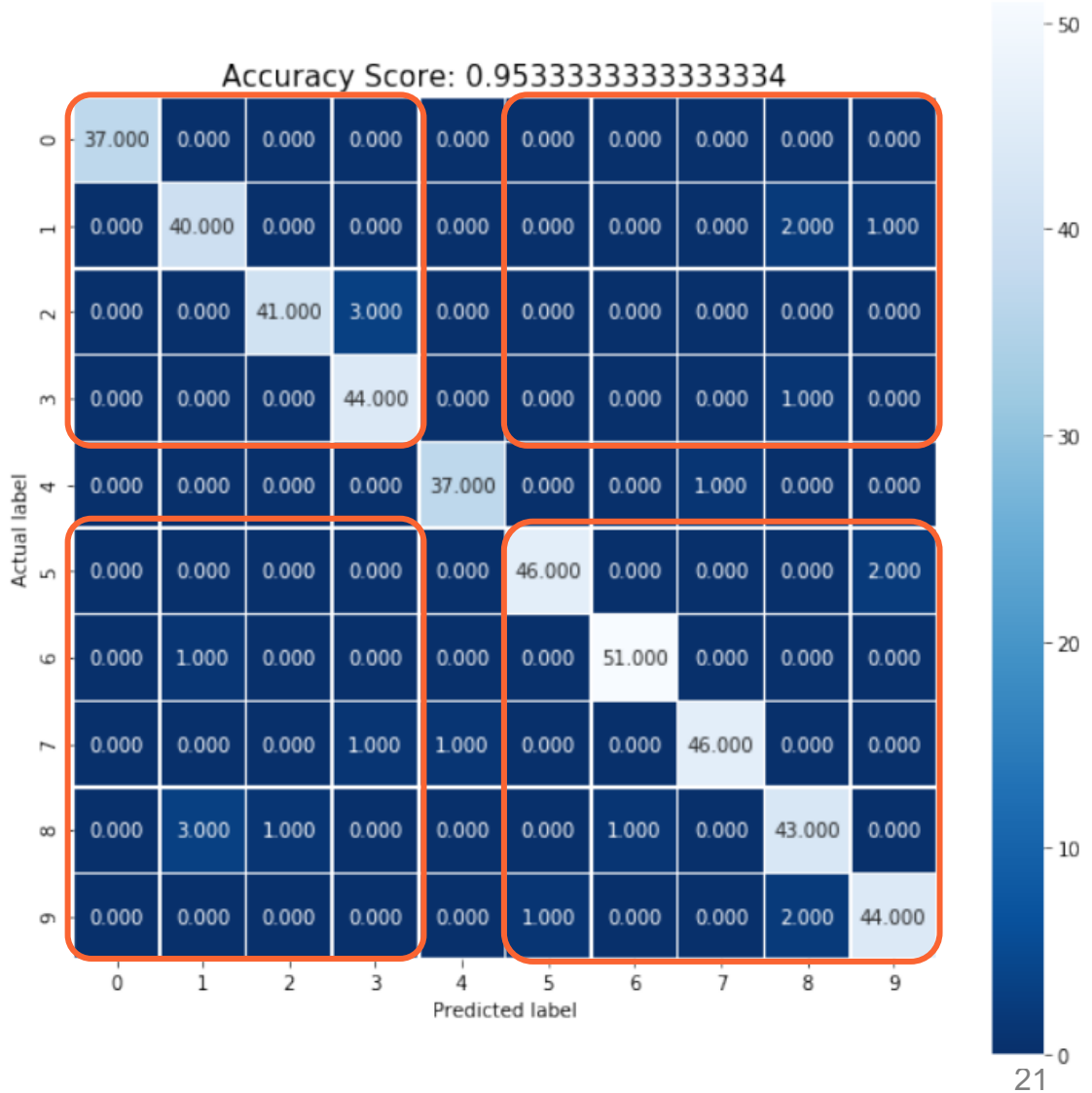
Example: class ‘4’
Red indicates FN



Confusion matrix

- The same goes for multiclass classification
- Calculate TP/TN/FP/FN **per class**
- You create an overal idea of your model by calculating an average
- Python can do both for us :D

Example: class ‘4’
Red indicates TN



Accuracy

- Measures the proportion of correct predictions over all predictions.
- Between 0 and 1
- Higher is better
- “How correct are my predictions?”

$$Accuracy = \frac{\#correct\ predictions}{\#total\ predictions}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

- Measures how often the same prediction will be made, given more or less the same independent variables.
- Higher is better
- “How consistent are my predictions?”

$$Precision = \frac{TP}{TP + FP}$$



Accuracy vs precision

Let’s pauze for a moment to visualize the difference between accuracy and precision



Low accuracy
Low precision



Low accuracy
High precision



High accuracy
Low precision



High accuracy
High precision


OXI
CLEAN
OXI
CLEAN
OXI
CLEAN
OXI
CLEAN
OXI
CLEAN
OXI
CLEAN

Recall

- Also called the **True Positive Rate** (TPR)
- Proportion of actual positives correctly identified
- Focuses on minimizing false negatives
  - Important in medical diagnoses, …
- “Of all positive values, how many did my model recognize?”

$$Recall = \frac{TP}{TP + FN}$$

Specificity

- Also called the **True Negative Rate** (TNR)
- Proportion of actual negatives correctly identified
- Focuses on minimizing false positives
  - Important in fraud detection, …
- “Of all negative values, how many did my model recognize?”

$$Recall = \frac{TN}{TN + FP}$$

F1 Score

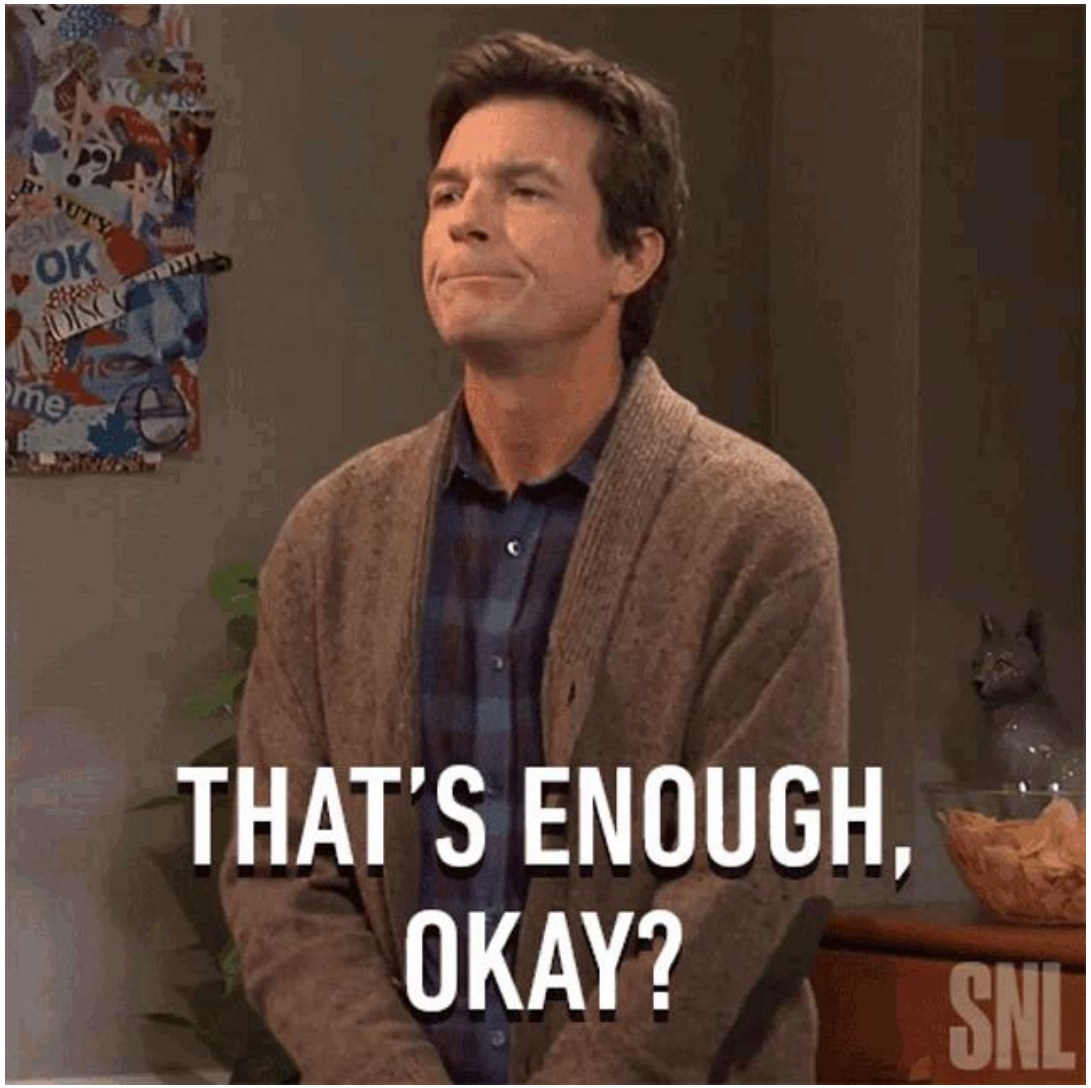
- Harmonic mean of precision and recall
- Gives a more balanced overview than accuracy

$$F1\ Score = 2\ \frac{Precision\ *Recall}{Precision + Recall}$$



Further metrics

- Is this all? Of course not!
- There are many more metrics such as **False Positive Rate** (FPR), **False Negative Rate** (FNR), **ROC**, …
- However, these fall out of the scope of Data Science Fundamentals.
- You will be seeing them again in Data Science Advanced ;)



THOMAS MORE

Let's jump into action

Excercises

First, let's explore this together

- Go to canvas and download the notebook "DSF 04 – Logistic Regression Notebook"
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Follow along!



Now it's up to you!

- Go to canvas and download the notebook "DSF 04 - Logistic Regression Exercise"
- Also download the csv file called "Diabetes.csv"
- Place them in the same directory
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Go through the notebook and make the excercises

I don't bite, ask me any questions you have!

